Inorganic Cumulative Examination

April 3, 2009

1. Henry Taube has made some very significant contributions to inorganic chemistry and his work has been recognized by many awards including the Nobel Prize. Among the specific reactions mentioned in that award is the reaction of Cr(II) with $Co(NH_3)_5Cl^{2+}$. In terms of mechanisms of electron transfer what very significant feature did it demonstrate and how was it demonstrated?

2. Later in his career, the Creutz and Taube complex ($[Ru(NH_3)_5]_2(C_4H_4N_2)^{5+}$) was prepared which demonstrated a nearly identical feature but in a very different way. Describe the significance of this complex in the development of electron transfer theories.

3. $Fe(phen)_3^{3+}$ is a very strong oxidant that reactions by a different mechanism than the oxidant in question 1. Explain the difference, explain why we can be so certain of the mechanism and how such reactions ultimately contributed to another Nobel Prize.

4. Permanganate ion is yet another extremely strong oxidant that reactions by way of a third mechanism. Describe the mechanism, explain how we can be so certain of the mechanism and explain why such reactions were ignored in the development of the electron transfer theories.